

University of Mosul
College of Computer Sciences and
Mathematics
Computer Science Dept. /First Class

Principles of Computer
Organization
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Lec. 5

presentation
notes

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8086 microprocessor internal registers



8086 consists of 13 register , All registers of size 16-bits

- ❑ Data registers (AX, BX, CX, and DX).
- ❑ *Segment registers* (CS, DS, SS, and ES).
- ❑ Instruction pointer (IP).
- ❑ Index registers (SI and DI).
- ❑ Pointer registers (BP and SP).
- ❑ Status register (SR) or (Flag register).

General Purpose Registers (data register) (Ax , BX , CX , DX)



AX register (Accumulator)

used for all input/output operations some string operation and Arithmetic operations.

BX register (base register)

used as an index to extend addressing it's also used for computation

CX register (count register)

used for controlling the number of times a loop is repeated contains the value by which bits are shifted it's also used for computations.

DX register (data register)

Used for input/output operations. It use for multiply and divide operations

General Purpose Registers (data register) (Ax , BX , CX , DX)



High byte low byte



↓
AX = 25F8
AL = F8
AH = 25

BX = A5D
BH = 0A
BL = 5D

AX	
AH	AL
BX	
BH	BL
CX	
CH	CL
DX	
DH	DL

A: Accumulator

B: Base reg.

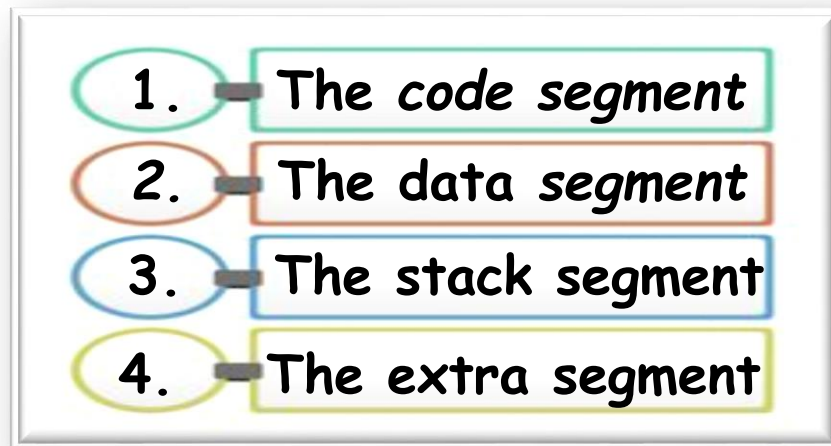
C: Counter reg.

D: Data reg.

- ✚ Each register can be **accessed as a byte or a word.**
- ✚ **The left most byte is a high-order 8 bits of register and the right most byte is the low-order 8bits.**

Segment Registers and Memory Segmentation

- Even though the 8086 has a 1Mbyte address space, **not all this memory can be active at one time.**
- The 1Mbytes of memory can be **partitioned into 64Kbyte (65,536) segments.**
- **A segment: represents an independently addressable unit of memory consisting of 64K consecutive byte-wide storage locations.**
- Each segment is assigned **a base address** that identifies its **starting point (its lowest address byte-storage location).**
- Only four segments can be active at a time



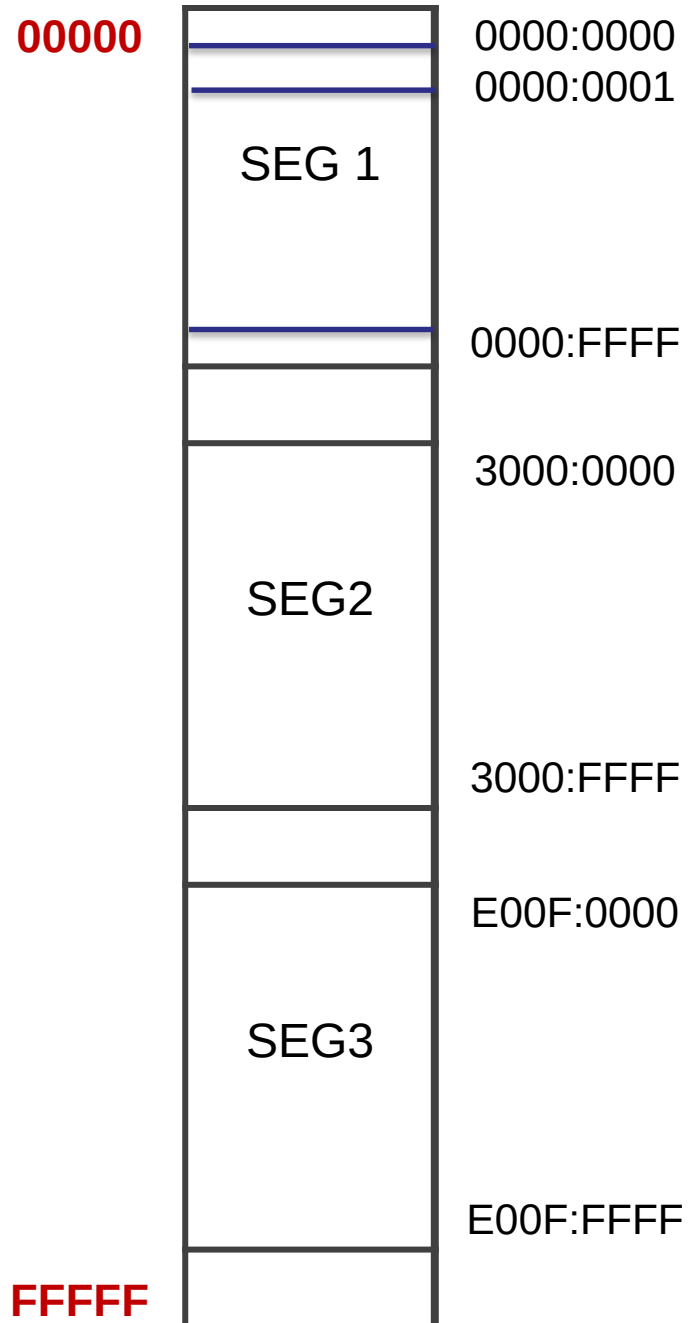
Size of segment=64kbyte

$$2^6 * 2^{10} = 2^{16}$$

Size of address in segment = 16 bits
0000

FFFF

Memory Segmentation



Segment Registers and Memory Segmentation



- ☞ Code Segment (CS) : holds the program instruction codes.
- ☞ Data Segment (DS) : stores data for the program.
- ☞ Stack Segment (SS) : used to store interrupt and subroutine return address.
- ☞ Extra segment (ES) : is an extra data segment.

The segments of memory that are active are identified by the values of addresses held in the 8086's four internal segment registers :

Code segment register (CS)

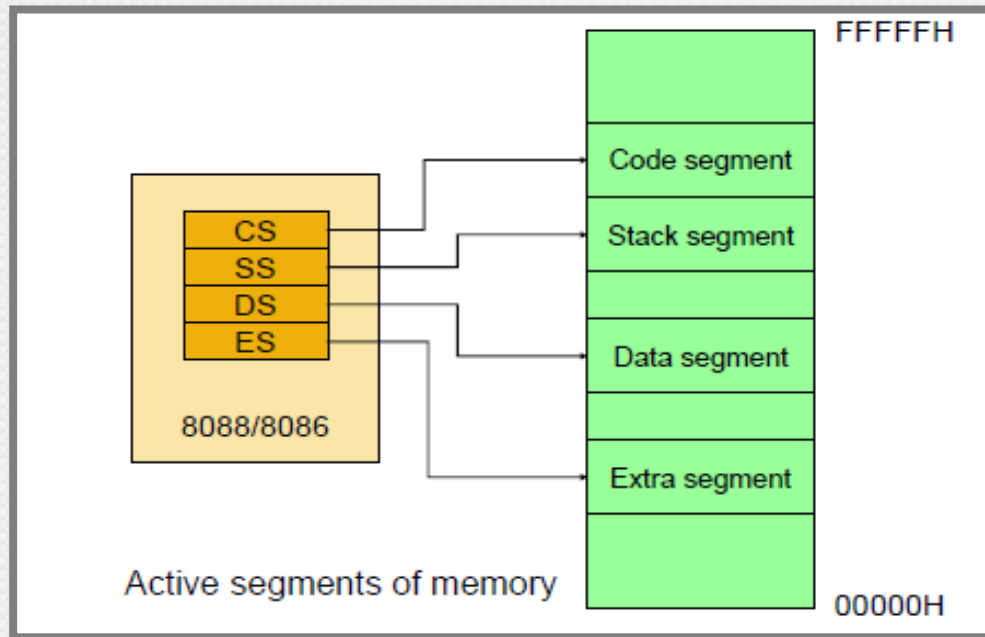
Data Segment Register (DS)

Stack Segment Register (SS)

Extra Segment Register (ES)

Each of these registers contains a **16-bit base address** that points to **the lowest addressed byte of the segment in memory**.

Segment Registers and Memory Segmentation



4 seg. active
 $4 \times 64\text{kbyte}$
 $= 256\text{kbyte}$

Four segments give a maximum of 256Kbytes of active memory.
{ 64Kbytes are for code (program storage), 64Kbytes are for a stack, and 128Kbytes are for data storage }

Logical and Physical addresses



Three types of addresses :

Physical Address

20 bits actually
put on the
address bus

Range: 00000 →
FFFFF

Offset Address

A location within
a 64K byte
segment

Range: 0000 → FFFF

Logical Address

Consist of a
segment value
and offset
address

SEGMENT VALUE: OFFSET ADDRESS

- ❖ The segment base address and offset address are **16 bit quantities**.
- ❖ This is because all register and memory locations used in address calculations are always 16 bits long. However, the physical addresses that are used to access memory are 20 bits in length.

Pointer and Index registers

Instruction pointer (IP):

Is a 16 bit register contains the offset of the next instruction to be fetched from the current code segment instead of the actual address. Every time an instruction is fetch 88/86 updates the value of IP by incrementing it. (CS:IP)

Pointer and Index registers:

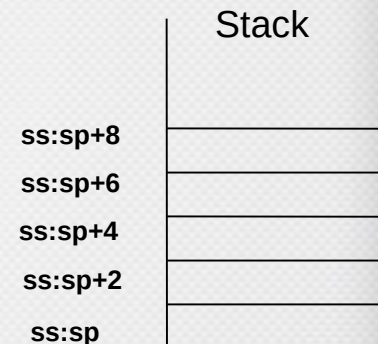
Pointer and index group is 16 bits Registers (**you cannot access the low or high bytes alone**). These Registers are used as memory pointer.

Pointer registers

Stack pointers (SP): represent the offset of the next stack location that is to be accessed.

(SS: SP): result a 20 bit address points to the top of the stack.

Base pointer (BP): The base pointer facilitates referencing of parameters. (Data and addresses passed via stack).



Pointer and Index registers



Index register

Source index (SI): is required for some string operations in this context the SI are associated with DS register. **(DS:SI)**

Destination index (DI): is required for some string operations in this context the DI are associated with ES register. **(ES:DI)**

Example:

MOV SI,1000

MOV AH,[SI]

What are the contents of register AH after the previous instructions is executed?

AH= 17

DS:1000	17	←DS:SI
DS: 1001	3A	
DS: 1002	26	

Summary (Offset registers for various segments)



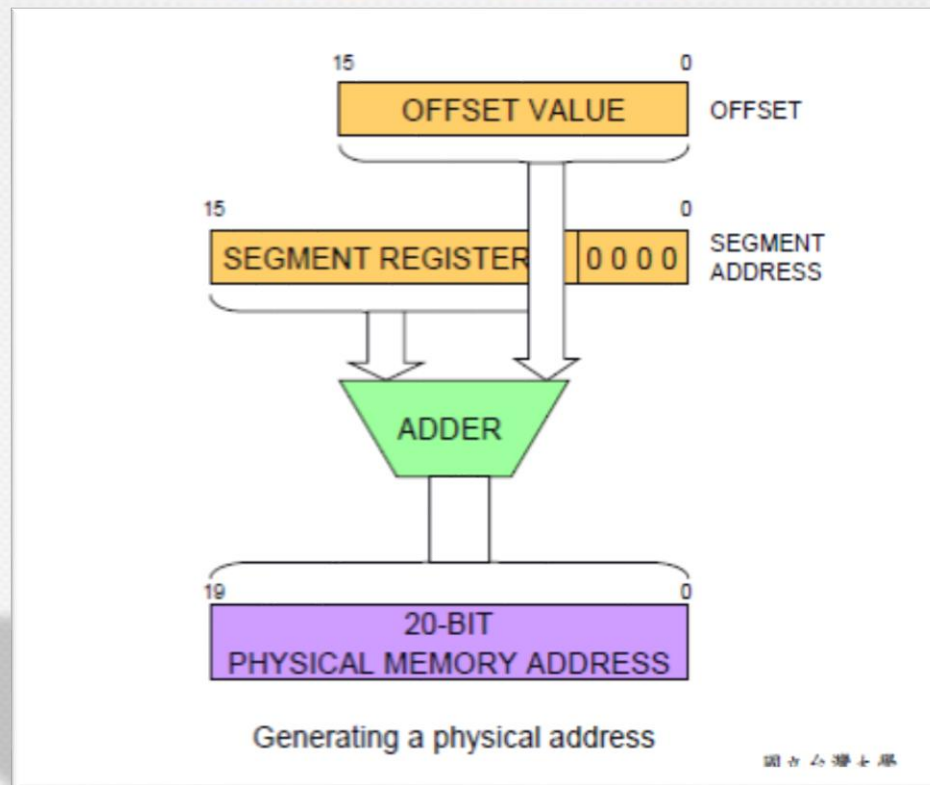
Segment Register	CS	DS	ES	SS
Offset Register(s)	IP	SI DI BX	DI SI BX	SP BP

CS:IP

DS: SI , DS:BX , DS:DI

SS:BP

Convert Logical address to Physical addresses



$$\text{Physical address (ph)} = \text{segment value} * 10 + \text{offset value}$$

Examples

Example1:

find the physical address for each of the following logical addresses:

1. 1000: 5020
2. 1400: 1020
3. E90F: 2302
4. 1302: 2009
5. 08F0: 0200

1/ SEGMENT VALUE = 1000 , OFFSET ADDRESS = 5020

PH= Segment value *10 + OFFSET

$$10000 + 5020 = 15020$$

10000

5020

15020

3/

segment value = E90F offset = 2302

$$E90F * 10 + 2302 = EB3F2$$

E90F0

2302

EB3F2

Example2:

If DS=2567 and SI = 2341 find:

1. The logical address.
2. The offset address.
3. The physical address.
4. The lower address in data segment
5. The upper address in data segment.

: الحل

1. DS : SI (2567:2341) LOGICAL ADDRESS
2. 2341
3. $PH = DS * 10 + SI$
 $= 2567 * 10 + 2341 = 25670 + 2341 = 279B1$
4. DS: 0000 (2567:0000)
5. DS:FFFF (2567:FFFF)

DS:0000

DS:FFFF



:
DS
:

Example3:

A code segment is to be located from physical address C1000 to D0FFF So, find the value that be loaded into CS register.

$$PH = SEG * 10 + OFFSET$$

$$C1000 = CS * 10 + 0000$$

$$C1000 = CS * 10$$

$$CS(CODE\ SEG.\ REG.) = C1000 / 10$$

CS:0000 ← C1000

D0FFF



:
CODE SEG.
: